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RESEARCH

Masked pose embedding for view and occlusion invariance in 2D pose analysis

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Abstract

Pose embeddings provide a practical way to improve viewpoint invariance and occlusion robustness in 2D pose analysis by transforming sparse landmark coordinates into representations that preserve underlying 3D pose semantics. Existing approaches often rely on contrastive or variational objectives, which require careful pair construction, large batches, or additional training constraints. In this paper, we propose masked pose embedding (MPE), a transformer-based autoencoder trained with a single masked-reconstruction objective tailored to pose landmarks. MPE combines two key components: (1) view-angle canonicalization, in which randomly rotated 3D landmarks are projected to 2D and decoded toward the original canonical pose, and (2) landmark-level masking with masked-only supervision to improve robustness to partial visibility. We evaluate MPE in two complementary settings. First, on SARA and Human3.6M, we analyze whether distances in the learned embedding space correlate with ground-truth 3D pose distances under both controlled perturbations and under natural, unperturbed data. Second, on Yoga-82, we assess real image-based pose retrieval using YOLOv8x-pose landmarks under multiple synthetic occlusion patterns. The results show that MPE more faithfully preserves 3D pose semantics than recent single-frame and transformer-based representation baselines and achieves stronger retrieval accuracy among the compared methods under both full and partial landmark observations. These findings suggest that MPE provides an effective and training-stable solution for view- and occlusion-robust 2D pose embedding, while remaining highly parameter-efficient compared with recent single-frame and temporal representation baselines.

Keywords Human pose estimation · Pose embedding · Masked reconstruction · View invariance · Occlusion robustness

1 Introduction

Human Pose Estimation (HPE) aims to localize articulated body joints from images or videos and serves as a fundamental component in a wide range of applications, including sports analytics [17, 43], healthcare and rehabilitation [8, 13, 18, 20], and surveillance [1, 21]. Although recent advances in 2D HPE have substantially improved landmark localization accuracy through regression- and heatmap-based formulations [6, 25, 38, 41], fine-grained pose analysis remains highly sensitive to camera viewpoint variation and landmark occlusion [35]. In practice, these factors can substantially distort 2D pose geometry even when the underlying 3D pose semantics remain similar, making reliable pose comparison and retrieval difficult.

A natural solution is to estimate or recover 3D pose structure. Indeed, 3D HPE can alleviate viewpoint dependence by explicitly modeling depth [11, 27, 39, 40]. However, 3D estimation remains affected by depth ambiguity, scale normalization issues, and increased computational cost, even with recent diffusion- and transformer-based approaches [7, 10, 33, 34, 45, 47]. For many downstream tasks, a more practical alternative is to learn a *pose embedding* that preserves 3D-aware pose similarity directly from 2D landmarks, without requiring full 3D reconstruction at inference time.

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Prior work has explored this direction using contrastive or variational learning. In particular, methods such as PrVIPE and V-VIPE [22, 24] improve robustness to viewpoint changes and occlusion through camera augmentation, joint masking, and latent-space regularization. While effective, these approaches typically depend on triplet or contrastive objectives, careful positive–negative pair construction, or additional variational constraints. Such formulations increase training complexity and may be sensitive to design choices such as batch composition, margin tuning, and objective balancing.

In parallel, masked reconstruction has emerged as a strong self-supervised learning principle for dense inputs, as demonstrated by Masked Autoencoders (MAE) [12]. However, directly transferring this paradigm to human pose landmarks is non-trivial. Unlike image patches, pose landmarks are sparse and semantically indexed, so each joint carries substantial structural meaning. As a result, both the masking strategy and the reconstruction target must be carefully designed. Simply reconstructing the same input view is insufficient when the goal is to learn an embedding that is robust to viewpoint changes and partial visibility.

To address these challenges, we propose Masked Pose Embedding (MPE), a transformer-based autoencoder that learns view- and occlusion-robust pose embeddings from sparse 2D landmarks using a single masked-reconstruction objective. MPE combines two key ideas. First, we apply *view-angle canonicalization*: 3D landmarks are randomly rotated, projected to 2D, and used as encoder input, while the decoder reconstructs the corresponding canonical unrotated pose. Second, we apply *landmark-level masking* and compute the loss only on masked joints, encouraging the model to infer missing pose structure rather than merely reproduce visible landmarks. This formulation directly promotes viewpoint invariance and occlusion robustness while avoiding pair mining, large-batch contrastive training, and variational balancing.

We evaluate MPE in two complementary ways. First, on the SARA dataset [31], we measure the Pearson correlation between pairwise distances in the learned embedding space and those in the ground-truth 3D pose space under 12 controlled conditions defined by viewpoint rotation and landmark masking. To further assess whether the learned embedding preserves 3D pose semantics on original Human3.6M samples, we additionally evaluate the same correlation protocol on Human3.6M [14] without artificial perturbations. Second, on the Yoga-82 dataset [42], we evaluate an image-based pose retrieval task using landmarks extracted by YOLOv8x-pose [16], including multiple synthetic occlusion settings. Across these evaluations, we compare MPE with representative baselines ranging from autoencoder-based methods and contrastive pose embeddings to recent single-frame view-invariant embedding and temporal skeleton representation learning approaches.

Our main contributions are as follows. (1) We propose Masked Pose Embedding (MPE), a transformer-based pose embedding framework whose main contribution is a pose-specific masked reconstruction objective that jointly couples *canonical-view reconstruction*, *landmark-level masking*, and *masked-only supervision* for sparse 2D pose landmarks. (2) We show that this simple reconstruction-based objective provides a practical and training-stable alternative to contrastive and variational pose embedding methods, without requiring explicit pair mining, large batches, or additional latent regularization. (3) We introduce a comprehensive evaluation protocol that examines pose embeddings from two complementary perspectives: controlled robustness analysis on SARA and natural-data correlation analysis on Human3.6M, together with real-image pose retrieval on Yoga-82 under multiple occlusion settings. (4) Through extensive experiments, we show that MPE preserves 3D-aware pose similarity more faithfully than representative baselines and achieves strong robustness to both viewpoint variation and partial occlusion, while remaining highly parameter-efficient.

2 Related work

We group prior work into four strands and emphasize how our Masked Pose Embedding (MPE) differs. In short, MPE is a *masked reconstruction* objective on sparse landmarks that (i) imposes view invariance by reconstructing a *canonical, unrotated* pose from randomly rotated inputs and (ii) imposes occlusion robustness by reconstructing *only* the masked joints. This avoids pair mining and large batches required by contrastive learning while keeping all supervision self-contained.

2.1 Masked reconstruction

Masked Autoencoders (MAE) showed that reconstructing randomly masked inputs is a strong self-supervised pretext on images with an asymmetric encoder–decoder design [12]. Variants extend masking to facial video [23], 3D point clouds [29], pose heatmaps for occlusion pretraining [48], target-focused interaction regions [46], and patchified skeleton sequences for motion synthesis [28]. However, these works typically operate on dense inputs (pixels/points/patches) or heatmaps where redundancy is high. In contrast, pose landmarks are sparse and semantically loaded—each joint matters—so masking ratios and targets must be adapted. Our MPE performs masking at the

landmark level and couples it with view augmentation: we rotate 3D landmarks, project them to 2D for the encoder, and train the decoder to reconstruct the *original, unrotated* 2D pose at masked indices. This differs from MAE-style targets (reconstructing the same-view input) and directly encodes view invariance while remaining lightweight.

2.2 Pose embeddings

Contrastive learning (e.g., SimCLR [5]) aligns augmented views but often requires large batches, careful temperature or margin tuning, and curated positive–negative pairs. In pose representation learning, Pr-VIPE [24] is a representative approach that learns view-invariant embeddings from 2D joint coordinates using camera augmentation, landmark masking, and triplet loss. Its extension, V-VIPE [22], further adopts a variational formulation in a canonical 3D coordinate space, explicitly targeting view-invariant pose embedding and supporting downstream tasks such as retrieval, classification, and 3D pose decoding. While V-VIPE provides a strong recent single-frame baseline for view-invariant pose representation, its primary objective is variational canonical embedding rather than masked reconstruction for explicit partial-landmark robustness. In contrast, MPE directly trains the model to reconstruct only the masked landmarks in the canonical view, thereby coupling view invariance and occlusion robustness within a single reconstruction objective.

Recent studies have also explored broader representation-learning settings. MotionBERT [49] learns general human motion representations from corrupted 2D skeleton sequences with a spatio-temporal transformer, while Uni-HPR [15] aligns RGB images, 2D poses, and 3D poses in a shared representation space. Although these methods can also be used for pose similarity, their primary design goals differ from ours. In contrast, we focus specifically on lightweight embedding learning from sparse 2D landmarks for view- and occlusion-robust pose similarity analysis and retrieval.

2.3 Multi-view 3D pose estimation

Methods that exploit calibrated multi-view imagery avoid label costs by cross-view consistency. SelfPose3d [36] jointly localizes persons and estimates 3D poses via synthetic supervision and differentiable projections, achieving results competitive with supervised baselines on Panoptic/Shelf/Campus. These approaches, however, presuppose multi-camera setups and known calibration, limiting applicability to monocular, in-the-wild scenarios. MPE is complementary: it requires only single-view 2D landmarks at inference, yet still encodes view invariance by training to invert random 3D rotations.

2.4 Occlusion invariance in pose similarity

Before learned embeddings, similarity often relied on limb lengths and joint angles [2, 9, 26, 30], which are sensitive to person-specific anatomy and camera viewpoint. Autoencoders have been used to learn skeletal features [3] and to fill missing joints under occlusion with denoising objectives [4]; sub-graph augmentations further improve partial-visibility robustness [44]. Other view-aware formulations map 2D poses to view-robust spaces with triplet losses [32, 37]. Our MPE unifies these desiderata in a single transformer autoencoder: (1) *view-angle augmentation* that trains reconstruction to a canonical pose, directly enforcing view invariance; and (2) *landmark-level masking* with a masked-only loss, which targets occlusion robustness without auxiliary classifiers or explicit pair/triplet mining.

3 Masked pose embedding

3.1 Overall pipeline

Figure 1 illustrates the proposed Masked Pose Embedding. MPE is based on a transformer autoencoder, but differs from conventional autoencoders in its use of view-angle augmentation and masked-only reconstruction. Firstly, view angle-based landmark augmentation is applied, which involves rotating the original 3D landmarks to capture diverse viewpoints. Secondly, masked landmark augmentation occludes specific portions of these augmented landmarks. These two augmentation strategies are seamlessly integrated into the Autoencoder framework, thereby significantly enhancing the model's overall robustness. Importantly, these components are not used as standalone augmentations; they are coupled with a canonical-view reconstruction target and masked-only loss so that the model learns a pose embedding that is explicitly robust to viewpoint changes and partial visibility.

3.2 View angle-based landmark augmentation

When representing 3D information within a 2D projection space, changes in viewpoint do not inherently alter the intrinsic content of the information but rather induce significant variations in its visual appearance. To address this, our proposed method incorporates an angle-based view landmark augmentation strategy designed to achieve view invariance.

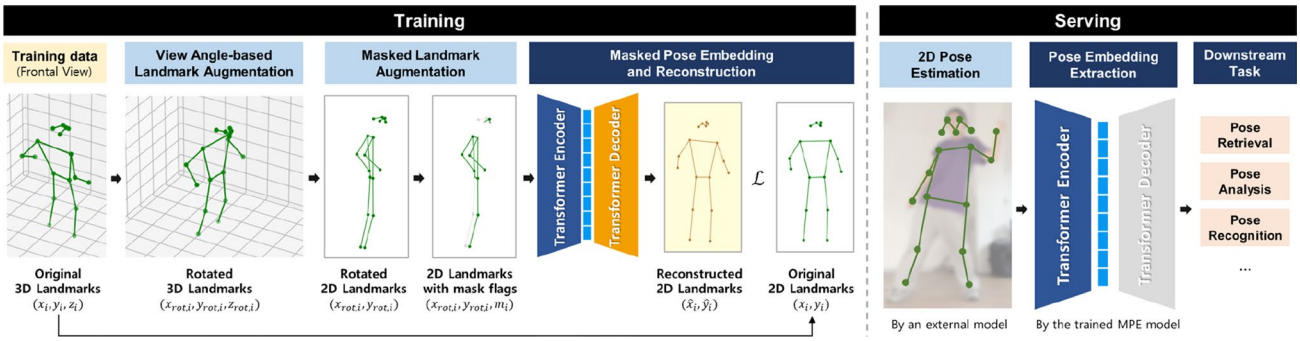


Fig. 1 Training and inference pipeline of the proposed Masked Pose Embedding (MPE). During training, rotated and partially masked 2D landmarks are used as input, and the model reconstructs the canonical unrotated pose at masked landmark positions. During inference, only the encoder is used to extract pose embeddings from estimated 2D landmarks

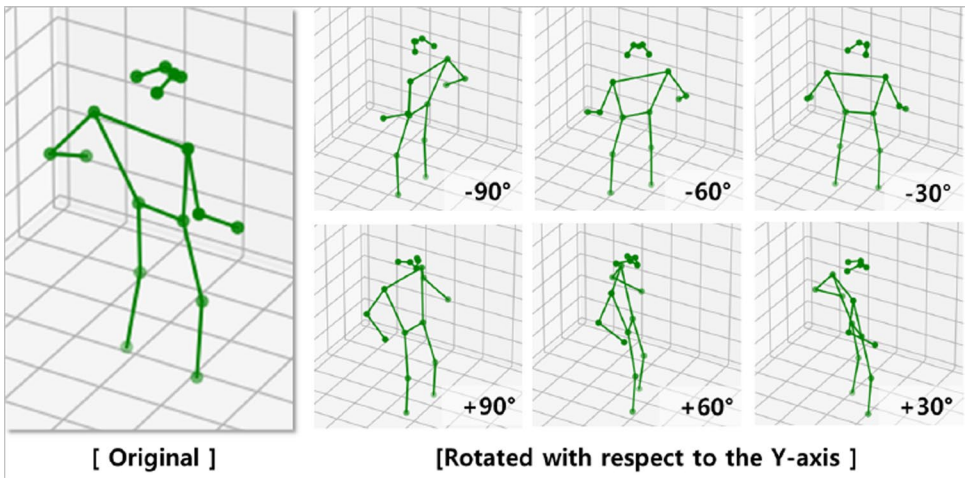


Fig. 2 Example of view-angle augmentation by y-axis rotation. The same 3D pose produces different 2D landmark configurations under different yaw angles while preserving the underlying pose semantics

For this purpose, a training dataset comprising 3D pose landmarks captured from a frontal view is required. During the training phase, upon receiving a 3D pose landmark, it undergoes arbitrary rotation along its X, Y, and Z axes. As illustrated in Fig. 2, the 3D coordinate data are rotated along these axes within predefined ranges. Subsequently, only the 2D coordinates of the rotated data are extracted and utilized as inputs to the model. The rotation of a 3D point (x, y, z) is mathematically performed using standard rotation matrices corresponding to each axis. The rotation matrices for rotations about the x, y, and z axes by angles θ_x, θ_y, and θ_z, respectively, are given by:

$$R_x(\theta_x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{bmatrix}$$

$$R_y(\theta_y) = \begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$$

$$R_z(\theta_z) = \begin{bmatrix} \cos\theta_z & -\sin\theta_z & 0 \\ \sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

To perform a combined rotation about the x, y, and z axes sequentially, the overall rotation matrix R is obtained by the product of the individual rotation matrices:

$$R = R_z(\theta_z)R_y(\theta_y)R_x(\theta_x)$$

The transformed coordinates (x', y', z') can be obtained by applying a rotation matrix R to the original coordinates (x, y, z) in 3D space:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = R \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

3.3 Masked landmark augmentation

Pose analysis frequently encounters occlusions, which can substantially degrade landmark reliability. These occlusions can arise from various sources: parts of the body may be obscured by external objects, or self-occlusion may occur due to the pose itself. For instance, in sports activities, a player's arm might be hidden behind their torso, or during everyday actions, one leg might obscure the other

when crossing. Although contemporary pose estimation techniques attempt to predict these occluded parts, the associated confidence values often remain low, leading to inaccurate predictions. This issue becomes critically important in precise applications, such as rehabilitation exercises, where accurate pose analysis is paramount.

To address these challenges, our approach builds upon the original Masked Autoencoder (MAE) method [12], which segments an input image into multiple patches and applies masking at the patch level. In contrast, our proposed method focuses specifically on pose landmarks. We receive pose landmarks as input and perform masking directly at the landmark level. Our method provides the encoder with both the coordinates of the pose landmarks and a binary vector indicating whether each specific landmark is masked. This masking vector is designed to randomly select landmarks for occlusion during training, thereby ensuring that a consistent proportion of landmarks is always masked. While the original MAE paper [12], which addressed image data, recommended a 75% masking ratio, our study on pose landmarks found that a 50% masking ratio yielded superior performance. For the masked landmarks, the encoder generates a latent representation. The decoder then utilizes this representation to reconstruct the original landmarks, including those that were initially masked.

3.4 MPE loss

Figure 3 illustrates the encoder-decoder architecture of the Masked Pose Embedding, which is based on transformers. The training process commences with an input sequence of rotated landmarks, represented as a series of (x_i, y_i) pairs for each individual landmark l. Specifically, the ordered list of original landmarks is denoted as $L = \{l_1, l_2, \ldots, l_n\}$, and the corresponding set of rotated landmarks is $L_{\text{rot}} = \{l_{1,\text{rot}}, l_{2,\text{rot}}, \ldots, l_{n,\text{rot}}\}$. A subset of landmarks within this L_{rot} sequence is subsequently masked to form a new masked sequence, L'_{rot}, where the occluded landmarks are replaced by a dedicated `MASK` token:

$$L'_{\text{rot}} = \{l_{1,\text{rot}}, l_{2,\text{rot}}, \texttt{MASK}, l_{4,\text{rot}}, \ldots, l_{n,\text{rot}}\}$$

To maintain the spatial relationships inherent in the pose data, positional embeddings are added to the projected landmark coordinates. This operation can be mathematically represented as:

$$Z = \text{Emb}(L'_{\text{rot}}) + \text{PosEnc}(L'_{\text{rot}})$$

where $\text{Emb}(L'_{\text{rot}})$ denotes the embeddings of the input sequence, and $\text{PosEnc}(L'_{\text{rot}})$ represents the positional encodings.

The resultant embeddings are then processed by the transformer encoder, which effectively employs masked tokens to manage missing or occluded data points. This encoder is composed of multiple self-attention layers and feedforward networks. Each self-attention layer computes attention using the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

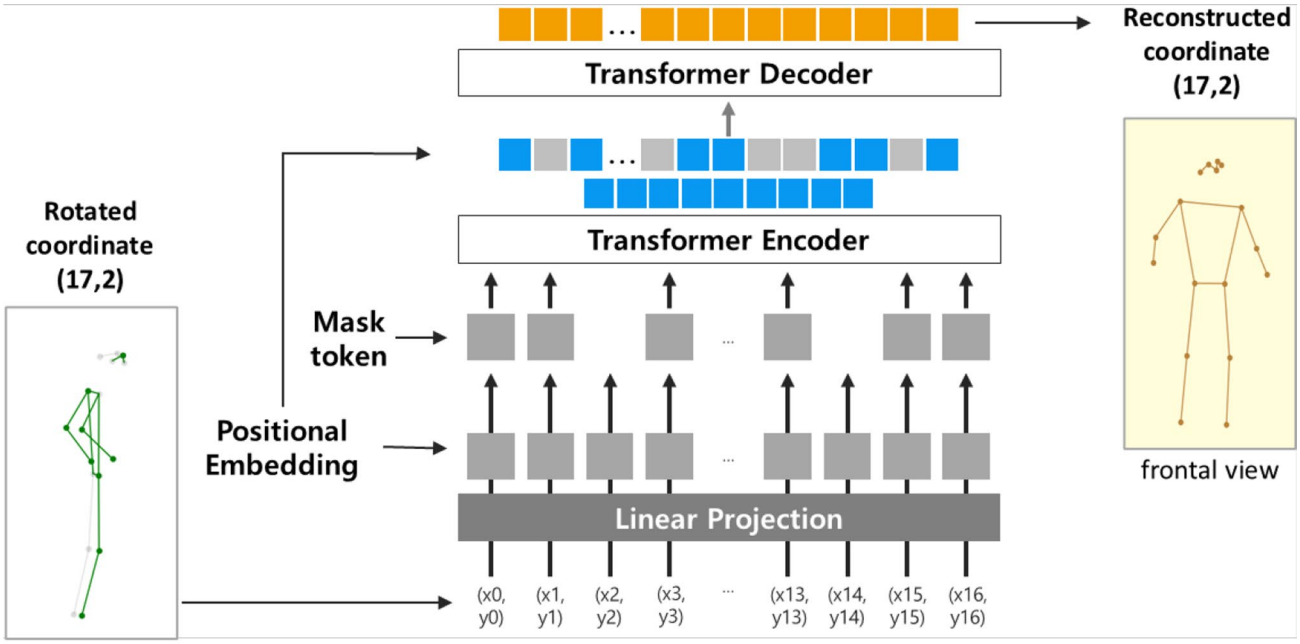


Fig. 3 Architecture of the proposed Masked Pose Embedding (MPE)

where $Q = K = V = Z$ and d_k represents the dimension of the embeddings. The final output obtained from the encoder after processing through several layers is denoted as H.

Subsequently, the encoded output H is fed into the transformer decoder. The decoder's primary function is to reconstruct the original landmarks by predicting the values of the masked tokens. The output of this decoder, denoted as Y, is calculated as:

$$Y = \text{Decoder}(H)$$

The decoder also consists of multiple attention layers and feedforward networks, structurally similar to those within the encoder.

To quantify the accuracy of the predictions, the Mean Squared Error (MSE) loss function is employed. Distinct from the original MAE, which typically uses the identical input fed to the encoder as the reconstruction target, our approach compares the predicted values Y with the original landmarks L (prior to any rotation), specifically focusing on the masked positions:

$$\mathcal{L}_{\text{MPE}} = \frac{1}{N} \sum_{i \in \mathcal{M}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

Here, (x_i, y_i) represents the actual coordinate values of the original landmark l_i (before masking and rotation for the target), while $(\hat{x}_i, \hat{y}_i)$ denote the corresponding predicted values, where $i \in \mathcal{M}$ indexes the masked landmark positions (i.e., $\mathcal{M}$ is the set of masked positions). This approach ensures that the model learns to accurately reconstruct the missing components of the input sequence, thereby enhancing its robustness against occlusions and variations in viewpoints.

This design is a key distinction from standard MAE-style reconstruction, since the model is not trained to reproduce the same observed input but to recover canonical pose structure specifically at the masked landmark positions. It also differs from conventional pose augmentation: the rotated view is not merely used to increase input diversity, but is explicitly paired with a canonical reconstruction target so that the embedding is encouraged to discard viewpoint-specific distortions while preserving pose semantics.

4 Analysis of masked pose embedding space

4.1 Methodology

We analyze the learned pose embedding space by measuring the Pearson correlation between pairwise L_2 distances in the embedding space and those in the ground-truth 3D pose space. A higher correlation indicates that the embedding better preserves underlying 3D pose semantics from 2D landmark inputs.

To assess both robustness under controlled perturbations and performance under natural pose distributions, we consider two evaluation settings. First, using the SARA dataset [31], we perform a controlled evaluation under 12 conditions formed by combinations of yaw rotation (0°, 30°, 60°, and 90°) and landmark masking ratios (0%, 25%, and 50%). This setting is designed to examine the stability of each embedding under explicit viewpoint changes and partial occlusions.

Second, using the Human3.6M dataset [14], we evaluate the same models on the original data without applying artificial rotation or landmark masking. Unlike the SARA-based controlled setting, this evaluation is intended to assess whether the learned embedding preserves 3D pose semantics under natural pose distributions.

In both settings, the evaluation protocol is based on pairwise comparison. For a given pose frame f_1, another frame f_2 is sampled, and we compute the Euclidean distance in the ground-truth 3D space as well as the corresponding distance in the learned embedding space. The Pearson correlation between these two distances is then used as the final metric.

4.2 Dataset

For the controlled evaluation, we used the Synthetic Actors and Real Actions (SARA) dataset [31]. The SARA dataset consists of 17 3D landmarks and includes 18 characters performing actions from four categories: Combat, Adventure, Sports, and Dance. Each motion sequence contains at least 32 frames. To ensure consistency across samples, the 3D landmark coordinates were normalized by centering the pelvis at the origin and scaling the values to the range of $[-1, 1]$. The dataset is divided into 455,028 training frames from 277 actions and 35,647 test frames from 23 actions.

For the correlation analysis on SARA, all 35,647 test frames were used. For each frame f_1, another frame f_2 was randomly sampled, and the Euclidean distance between their ground-truth 3D coordinates was computed. To evaluate robustness under controlled perturbations, the landmarks of f_2 were subjected to masking at rates of 0%, 25%, and 50%, and were simultaneously rotated around the y-axis by 0°, 30°, 60°, and 90°.

For the additional evaluation under natural pose distributions, we used the Human3.6M dataset [14], a large-scale benchmark for 3D human pose estimation. Human3.6M contains approximately 3.6 million 3D human poses and corresponding images, captured from 11 subjects using 4 calibrated cameras. In this study, we used 1,559,752 training

frames and 566,920 test frames in the original Human3.6M samples. As in the SARA experiments, the 3D landmark coordinates were normalized by centering the pelvis at the origin and scaling the values to the range of $[-1, 1]$. We then followed the same pairwise correlation protocol described in the Methodology subsection, computing the Pearson correlation between pairwise distances in the learned embedding space and those in the ground-truth 3D pose space.

4.3 Training details

Unless otherwise stated, the same training protocol was used for both the SARA- and Human3.6M-based experiments. The inputs to the encoder of the MPE model were augmented with both masked landmark augmentation and view angle-based landmark augmentation during training.

For masked landmark augmentation, a predefined proportion of landmark coordinates was randomly masked. Although the original MAE study [12] suggested a masking ratio of 75% for image data, our preliminary experiments on pose landmark data showed that a 50% masking ratio yielded better performance, and this setting was therefore used in the main experiments.

For view angle-based landmark augmentation, the X- and Z-axes were rotated by a random angle within $\pm 15^{\circ}$, while the Y-axis was rotated by a randomly selected angle from $\{-90^{\circ}, -60^{\circ}, -30^{\circ}, 0^{\circ}, 30^{\circ}, 60^{\circ}, 90^{\circ}\}$. The original, unrotated X and Y coordinates of the landmarks were used as the reconstruction target for the decoder.

The MPE model employed a six-layer transformer encoder, where each layer used four attention heads and an embedding dimension of 128. The decoder was designed as a lightweight two-layer transformer with four attention heads and an embedding dimension of 64. Training was performed using the AdamW optimizer with an initial learning rate of 5×10^{-3}. We adopted a cosine learning rate schedule with a warm-up phase corresponding to 15% of the total training steps, and applied a weight decay of 1×10^{-4}. The model was trained with a batch size of 32 for 10 epochs using the Mean Squared Error (MSE) loss.

4.4 Baselines

We compared the proposed method with five representative baselines: Pr-VIPE, V-VIPE, MotionBERT, AE, and VAE.

Pr-VIPE [24]: Pr-VIPE is designed to learn view-invariant and occlusion-robust embeddings directly from 2D joint coordinates. View invariance is achieved through camera augmentation using multi-view datasets and random 3D coordinate rotations, while occlusion robustness is encouraged by randomly masking coordinates and providing the corresponding mask vectors during training. The model is optimized with a triplet-loss formulation [32], where positive pairs are constructed from rotated anchor samples and negative pairs are selected based on large Mean Per Joint Position Error (MPJPE) distances from the anchor [14]. Our implementation followed the methodological details described in [24]. We tested mask ratios of 20% and 50% in combination with rotation augmentation, using triplet margins of 2 and 3.

V-VIPE [22]: V-VIPE was used as a recent single-frame, view-invariant pose embedding baseline. It is particularly relevant to our setting because it maps 2D and 3D poses into a canonical pose embedding space and supports downstream tasks such as pose retrieval and classification. We used a self-implemented version based on the methodological description in the original paper.

In its original formulation, V-VIPE primarily targets viewpoint-invariant pose representation rather than explicit robustness to partial landmark occlusion. Therefore, we evaluated masking-augmented variants by applying landmark masking during training. This allows V-VIPE to be evaluated under training conditions better aligned with our occlusion-oriented protocol. We report masking-augmented V-VIPE variants with 25%, 50%, and 75% training masking ratios in the controlled correlation analysis, and use the best-performing variant for the main retrieval comparison.

MotionBERT [49]: We included MotionBERT as a recent strong baseline for skeleton-based representation learning. Although MotionBERT was originally proposed as a general human motion representation model rather than a pose-retrieval-specific embedding method, it is highly relevant to our setting for three reasons. First, like MPE, it learns geometry-aware representations directly from 2D skeletal observations. Second, it is built on a Transformer-based architecture, specifically the Dual-stream Spatio-temporal Transformer (DSTformer), making it a strong modern baseline for representation learning. Third, its official training code and pretrained models are publicly available, enabling reproducible evaluation under a transparent experimental setup. For a fair comparison, MotionBERT was trained using the same batch size, optimizer, learning-rate scheduler, and number of epochs as the other methods. Its learning rate was selected through preliminary experiments, and the best-performing setting was used for evaluation. For masked evaluation inputs, we first reconstructed the missing landmarks using MotionBERT and then extracted pose embeddings from the reconstructed poses, rather than embedding the masked inputs directly.

Nevertheless, MotionBERT is not a directly matched single-frame pose embedding baseline, because it was originally designed for temporal skeleton representation learning. We therefore treat MotionBERT as a complementary recent representation-learning baseline and include V-VIPE

as the more directly comparable single-frame view-invariant pose embedding baseline.

Autoencoder (AE) [4, 19]: We adopted an autoencoder structure analogous to those used in prior work. To encourage robustness to viewpoint changes and occlusion, we applied random landmark masking together with rotation augmentation during training. The model was trained to minimize the Mean Squared Error (MSE) between the reconstructed output and the target landmark coordinates. We evaluated several configurations, including 50% masking with rotation, 50% masking only, and rotation only without masking.

Variational Autoencoder (VAE) [4]: Following the structure described by Carissimi et al. [4], we employed a VAE baseline. This model used the same augmentation settings as the AE baseline, namely random landmark masking and rotation augmentation. As with AE, the model was trained using an MSE reconstruction objective on the target landmark coordinates. We evaluated the same configurations as for AE: 50% masking with rotation, 50% masking only, and rotation only without masking.

For all methods, we used the same correlation-based evaluation metric. The evaluation protocol differed depending on the dataset. On SARA, controlled landmark masking and viewpoint rotation were applied at test time to assess robustness under explicit occlusion and view perturbations. On Human3.6M, all methods were evaluated on the original unperturbed samples to examine whether the learned embeddings preserve 3D pose semantics under natural pose distributions.

4.5 Results

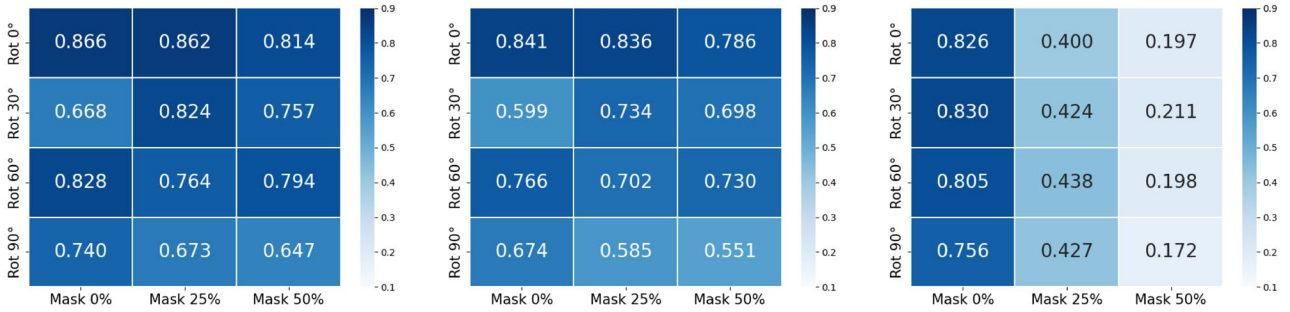
To identify the best configuration for each method, we first conducted an ablation study. Detailed ablation results, including analyses of the training objective, masking ratio, rotation augmentation, and architectural choices, are provided in the Appendix B and C.

As shown in Fig. 4, among all compared methods, the proposed MPE achieved the strongest and most consistent performance across the 12 evaluation conditions. In particular, MPE trained with a 50% masking ratio showed the best overall behavior, with correlations ranging from 0.647 to 0.866, and consistently outperformed the 75% masking variant. This result indicates that, for sparse and semantically indexed pose landmarks, a moderate masking ratio is more effective than aggressive masking. Although a 75% masking ratio is often effective in image-based MAE settings, it appears to remove too much structural information in the pose domain, making reconstruction and representation learning less stable.

MotionBERT exhibited a clear trade-off between viewpoint robustness and occlusion robustness. Under fully observed conditions (0% masking), it maintained relatively strong correlations across different viewpoints, and in some rotation settings it was even comparable to or better than the proposed MPE. This suggests that MotionBERT can capture viewpoint-robust global pose structure effectively when the input pose is fully available. However, once landmark masking was introduced, its performance deteriorated sharply, especially at 25% and 50% masking. Notably, this degradation was observed even though masked inputs were first reconstructed by MotionBERT before extracting the final pose embedding. Therefore, although MotionBERT can be competitive under clean, unoccluded inputs, it is substantially less tolerant to partial landmark removal than MPE in the present evaluation protocol. This contrast is particularly notable given that MotionBERT is a much larger model than MPE (42.47M vs. 1.98M parameters), yet MPE achieves clearly more stable performance under simultaneous viewpoint change and occlusion.

The V-VIPE baseline provides a more direct comparison with recent single-frame, view-invariant pose embedding methods. Because V-VIPE was not originally designed as an occlusion-specific embedding method, we evaluated masking-augmented V-VIPE variants under our occlusion-oriented protocol. Specifically, we compared training masking ratios of 25%, 50%, and 75% in the controlled SARA evaluation. Among these variants, training with a 50% masking ratio produced the strongest average correlation, but it still remained below MPE across the controlled viewpoint and occlusion conditions. This result suggests that simply adding masking augmentation to a variational view-invariant embedding framework is not sufficient to match the robustness obtained by MPE's canonical-view masked reconstruction objective.

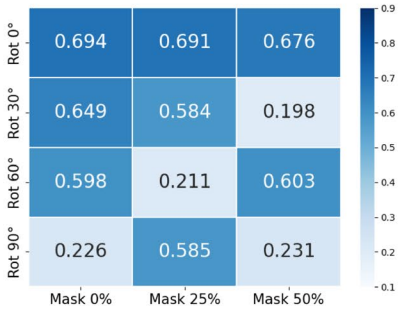
The remaining baselines also showed lower and less stable correlations than MPE. AE achieved moderate performance overall and remained relatively competitive under some low- and mid-difficulty settings, but its robustness weakened under larger viewpoint changes and heavier masking. Pr-VIPE showed strong sensitivity to the specific evaluation condition: while some view–occlusion combinations yielded reasonable correlations, others dropped substantially, indicating unstable behavior under severe perturbations. VAE also showed weaker overall alignment with 3D pose similarity and did not provide consistent robustness gains across the controlled settings. Taken together, these results indicate that the proposed masked reconstruction objective yields a more stable and semantically faithful pose embedding than both contrastive/variational pose embedding baselines and conventional autoencoding baselines under simultaneous viewpoint and occlusion perturbations.



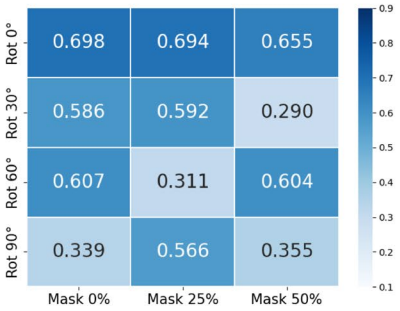
(a) MPE (50% masking)

(b) MPE (75% masking)

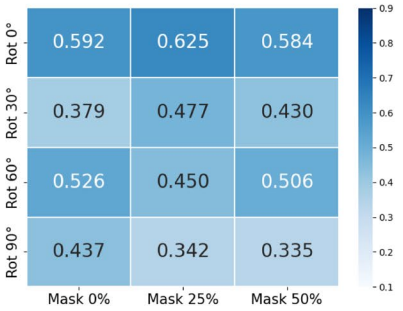
(c) MotionBERT (Original)



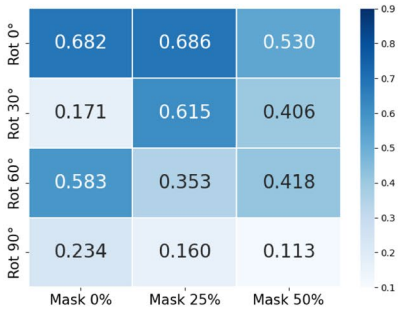
(d) Pr-VIPE (Original)



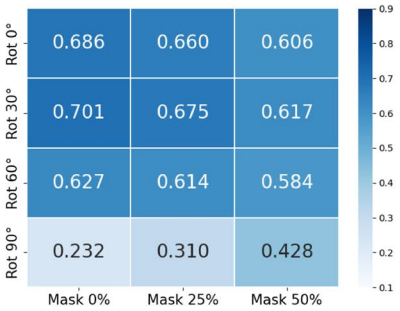
(e) Pr-VIPE (Best)



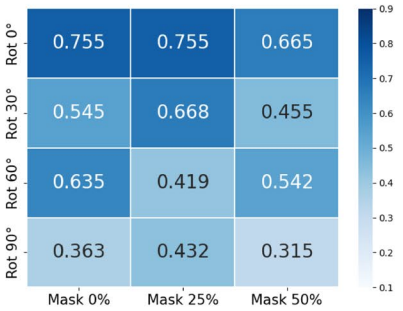
(f) VAE (Original)



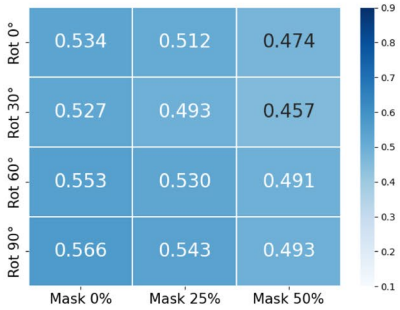
(g) VAE (Best)



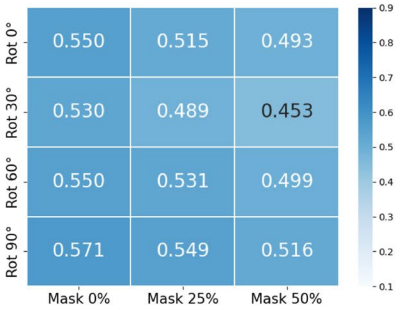
(h) AE (Original)



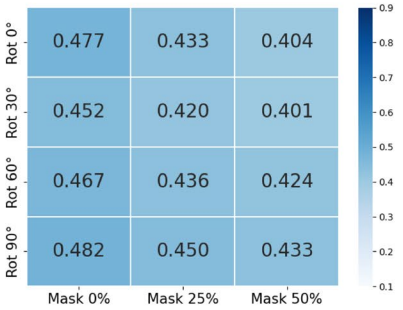
(i) AE (Best)



(j) V-VIPE (25% masking)



(k) V-VIPE (50% masking)



(l) V-VIPE (75% masking)

Fig. 4 Correlation analysis under controlled viewpoint and occlusion conditions. Each heatmap shows the Pearson correlation between embedding-space distances and ground-truth 3D pose distances across yaw rotations and test-time masking ratios. Higher values indicate better preservation of 3D pose similarity

In particular, although MotionBERT can be competitive in some unmasked rotation conditions, MPE provides clearly stronger robustness once occlusion is involved, while using far fewer parameters.

Table 1 presents the additional correlation results on Human3.6M. Unlike the controlled SARA setting, this evaluation was conducted on the original Human3.6M data without artificial rotation or landmark masking. Under this natural-data setting, MPE again achieved the highest correlation (0.838), outperforming MotionBERT (0.806), VAE (0.775), AE (0.730), Pr-VIPE (0.717), and the V-VIPE baseline (0.654). This result further supports that MPE's advantage is not limited to controlled synthetic perturbations on SARA, but also extends to natural Human3.6M pose distributions.

4.5.1 Ablation study

To better understand which components of MPE are responsible for the gains observed in the main experiments, we conducted a set of ablation studies on both the training design and the model architecture. These ablations are intended not only to optimize performance, but also to isolate the contribution of the main pipeline components of MPE, namely masked-only supervision, landmark masking ratio, view-angle-based canonicalization, and the lightweight transformer design. In this sense, the ablation study serves to clarify that the effectiveness of MPE does not arise from adopting masked modeling in a generic form, but from how it is specifically reformulated for sparse 2D pose landmarks.

We first examine the effect of the reconstruction target in Table 3, which compares *masked-only* supervision and *all-joint* supervision under different masking ratios. Overall, the masked-only setting is more stable across conditions and tends to yield stronger correlations, especially under larger viewpoint changes and heavier test-time occlusion. This result suggests that restricting the loss to masked landmarks encourages the model to infer missing pose structure more effectively, instead of allocating capacity to reconstructing already visible joints. Therefore, the improvement is not merely due to masking itself, but also to the way the reconstruction objective is defined.

Table 1 Correlation comparison on Human3.6M under natural, unperturbed pose distributions

Model	Corr. mean	SD	95% CI
MPE (Ours)	0.838	0.002	[0.834, 0.842]
MotionBERT	0.806	0.003	[0.800, 0.811]
Pr-VIPE	0.717	0.004	[0.710, 0.723]
VAE	0.775	0.003	[0.770, 0.779]
AE	0.730	0.003	[0.724, 0.736]
V-VIPE	0.654	0.005	[0.644, 0.662]

We next analyze the effect of the masking ratio. As shown in Table 4, 50% masking consistently outperforms both no masking and 75% masking across most viewpoint–occlusion settings. This finding indicates that, for sparse and semantically indexed pose landmarks, overly aggressive masking removes too much structural information and makes both reconstruction and representation learning less stable. In contrast, moderate masking provides a better balance between difficulty and recoverability, which is particularly important in pose data where each landmark carries substantial semantic meaning.

We further evaluate the contribution of view-angle-based canonicalization in Table 5, which compares MPE trained with and without rotation augmentation. The gains become more pronounced as the viewpoint becomes more challenging, especially under 60° and 90° rotations. This supports our central design choice of reconstructing the canonical unrotated pose from rotated inputs. Unlike standard MAE-style reconstruction, which aims to recover the same observed input, this formulation explicitly encourages the embedding to discard viewpoint-specific distortions and preserve pose semantics that are stable across views.

Finally, we examine the role of the model architecture through the ablations in Tables 6, 7, 8, and 9. To avoid selecting the architecture based on a single favorable condition, we used the mean Pearson correlation across all 12 controlled SARA evaluation conditions as the aggregate selection criterion. Under this criterion, the selected configuration achieved the highest aggregate mean among the tested variants. Specifically, the selected configuration achieved an average correlation of 0.770 across the 12 conditions, compared with 0.684/0.721 for Enc=3/9, 0.767/0.608 for dim=64/256, 0.654/0.642 for head=2/8, and 0.712/0.734 for dec=1/4.

We note that some alternative configurations outperform the selected configuration under specific individual conditions. In particular, dim=64 is a competitive compact alternative and performs better in several conditions. However, the selected dim=128 configuration provides the best aggregate mean under our predefined criterion.

Therefore, we interpret the selected architecture as the best aggregate setting among the tested configurations, rather than as a universally optimal architecture for every individual evaluation condition. These results also indicate that the gains of MPE arise from the combination of the proposed training objective and pipeline design rather than from model size alone.

These ablations support the main claim of this work: the contribution of MPE lies in a pose-specific reformulation of masked reconstruction, where landmark-level masking, masked-only supervision, and canonical-view

reconstruction jointly produce a more stable and semantically faithful embedding under viewpoint variation and partial occlusion.

5 Pose retrieval evaluation

5.1 Methodology

We evaluated the proposed embedding on a downstream *pose retrieval* task to assess whether the learned representation preserves semantically meaningful pose similarity in real image-based scenarios. In this experiment, we compared MPE with five baseline embedding methods: V-VIPE, MotionBERT, Pr-VIPE, VAE, and AE. For VAE, AE, Pr-VIPE, and V-VIPE, we used the best-performing configurations identified in the embedding-space correlation analysis. For V-VIPE, this corresponds to the best-performing masking-augmented variant selected from the controlled correlation ablation, while the full masking-ratio ablation is reported in the appendix. For MotionBERT, we used its standard representation setup and evaluated it as a recent strong baseline for human pose representation learning.

Given a query image from the Yoga-82 test set, we first extracted 2D human landmarks using YOLOv8x-pose [16]. The extracted landmarks were then normalized using the same preprocessing applied during training. Each embedding model converted the query landmarks into a pose embedding, and retrieval was performed by ranking all remaining images in the dataset according to the Euclidean distance in the embedding space. The query image itself was excluded from the candidate set. For each query, we retrieved the top-K most similar poses, with $K \in \{1, 3, 5\}$.

To evaluate robustness to partial visibility, we additionally applied synthetic masking to the query landmarks at test time. In addition to the unmasked setting (*None*), we considered six occlusion patterns: masking the right arm and right leg together, masking only the right arm, masking only the right leg, masking the left arm and left leg together, masking only the left arm, and masking only the left leg. This protocol allows us to examine how well each embedding preserves pose similarity when only partial body information is available.

We report two retrieval metrics: *Precision@K* and *HitRate@K*. For a query i, let $\mathrm{rel}_i@K$ denote the number of relevant items among the top-K retrieved results. Then,

$$\mathrm{Precision@K} = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathrm{rel}_i@K}{K},$$

and

$$\mathrm{HitRate@K} = \frac{1}{N}\sum_{i=1}^{N}1(\mathrm{rel}_i@K > 0),$$

where N is the number of query images. Higher values indicate better retrieval performance. A retrieved sample was considered relevant if it belonged to the same yoga pose category as the query.

5.2 Dataset

To evaluate retrieval performance in a practical downstream setting, we used the Yoga-82 dataset [42], which was not used during training. Yoga-82 contains in-the-wild images of 82 yoga poses captured under diverse backgrounds, viewpoints, lighting conditions, and frequent self-occlusions, making it suitable for assessing the robustness and generalization ability of pose embeddings in real-world scenarios.

The original Yoga-82 dataset provides image download links rather than a directly packaged image set. In our experiments, we first removed samples whose image links were no longer accessible. We then applied YOLOv8x-pose to extract 2D landmarks from the remaining images and excluded samples for which pose detection failed or produced unreliable results. This filtering step was necessary to ensure that the evaluation reflects the quality of the pose embeddings themselves, rather than failures of the upstream pose detector.

After preprocessing, a total of 14,203 images were retained for evaluation. The landmarks extracted by YOLOv8x-pose follow a structure similar to those used in the SARA dataset, but differ in coordinate range and reference center. Therefore, we applied the same normalization procedure used during training, along with an additional coordinate scaling step to map landmark values into the range of $[-1, 1]$.

5.3 Results

Table 2 summarizes the pose retrieval performance on Yoga-82 under different query masking conditions. Overall, the proposed MPE achieved the best retrieval performance across both *Precision@K* and *HitRate@K*, not only in the unmasked setting but also under all six partial-occlusion settings. These results indicate that the learned embedding generalizes effectively to real image-based retrieval scenarios and remains robust even when only incomplete landmark observations are available.

In the unmasked setting, MPE achieved the strongest overall performance, reaching 0.878 in *Precision@1*, 0.825 in *Precision@3*, and 0.798 in *Precision@5*, together with 0.878, 0.927, and 0.939 in *HitRate@1*, *HitRate@3*, and *HitRate@5*, respectively. These results outperform

Table 2 Pose retrieval performance on Yoga-82 under full and partially masked query poses. We report *Precision@K* and *HitRate@K* for $K = 1, 3, 5$ using Euclidean distance in the learned embedding space

Method	Masked	Precision			HitRate		
		@1	@3	@5	@1	@3	@5
MPE (ours)	None	.878	.825	.798	.878	.927	.939
	right arm+leg	.793	.789	.765	.793	.890	.915
	right arm	.817	.801	.783	.817	.890	.902
	right leg	.829	.801	.778	.829	.915	.951
	left arm+leg	.841	.801	.763	.841	.902	.927
	left arm	.866	.813	.780	.866	.902	.915
	left leg	.841	.813	.785	.841	.915	.927
V-VIPE	None	.728	.682	.656	.728	.828	.862
	right arm+leg	.690	.639	.609	.690	.806	.846
	right arm	.716	.668	.641	.716	.822	.856
	right leg	.705	.653	.624	.705	.812	.847
	left arm+leg	.686	.640	.612	.686	.801	.841
	left arm	.712	.666	.641	.712	.817	.853
	left leg	.712	.663	.636	.712	.819	.852
MotionBERT	None	.646	.587	.553	.646	.764	.809
	right arm+leg	.620	.557	.523	.620	.746	.794
	right arm	.583	.519	.484	.583	.722	.774
	right leg	.621	.557	.523	.621	.746	.794
	left arm+leg	.620	.557	.520	.620	.745	.791
	left arm	.631	.570	.537	.631	.752	.799
	left leg	.590	.530	.496	.590	.723	.772
Pr-VIPE	None	.646	.626	.622	.646	.744	.805
	right arm+leg	.744	.679	.651	.744	.805	.829
	right arm	.720	.659	.646	.720	.780	.829
	right leg	.671	.634	.617	.671	.744	.768
	left arm+leg	.671	.663	.641	.671	.780	.817
	left arm	.671	.646	.632	.671	.793	.817
	left leg	.720	.659	.646	.720	.780	.805
VAE	None	.573	.589	.576	.573	.707	.732
	right arm+leg	.573	.606	.598	.573	.744	.793
	right arm	.585	.557	.566	.585	.622	.683
	right leg	.634	.610	.598	.634	.744	.780
	left arm+leg	.598	.569	.571	.598	.671	.720
	left arm	.622	.577	.563	.622	.671	.732
	left leg	.561	.561	.556	.561	.695	.756
AE	None	.610	.581	.554	.610	.793	.829
	right arm+leg	.695	.659	.637	.695	.780	.805
	right arm	.695	.650	.637	.695	.793	.817
	right leg	.695	.683	.656	.695	.817	.854
	left arm+leg	.695	.687	.663	.695	.817	.841
	left arm	.744	.654	.646	.744	.793	.854
	left leg	.720	.659	.641	.720	.793	.841

all compared baselines, including the V-VIPE baseline (0.728/0.682/0.656 in *Precision@1/3/5*), MotionBERT (0.646/0.587/0.553), and Pr-VIPE (0.646/0.626/0.622). This shows that MPE is more effective at ranking semantically correct poses at the very top of the retrieval list even without artificial occlusion.

More importantly, MPE remained the strongest method under all partial-occlusion settings. Across the six masking patterns, *Precision@1* remained in the range of 0.793–0.866, while *HitRate@5* stayed between 0.902 and 0.951. The largest degradation was observed when the right arm and right leg were masked together, where *Precision@1* decreased to 0.793, whereas the most favorable masked setting was left-arm masking, where *Precision@1* remained as high as 0.866. Notably, even under all masked settings, *HitRate@5* remained above 0.90, indicating that MPE

preserves sufficient pose semantics to retrieve relevant candidates reliably despite partial landmark loss.

V-VIPE was the strongest competing method in the unmasked retrieval setting, achieving 0.728 in *Precision@1* and 0.862 in *HitRate@5*. This confirms that V-VIPE is a meaningful recent single-frame pose embedding baseline for retrieval-oriented evaluation. However, MPE still showed a clear advantage over V-VIPE, improving *Precision@1* from 0.728 to 0.878 in the unmasked setting and maintaining higher performance under all six partial-occlusion conditions. Under masked query conditions, V-VIPE's *Precision@1* ranged from 0.686 to 0.716, whereas MPE remained in the range of 0.793 to 0.866. These results indicate that MPE provides stronger occlusion robustness than a recent view-invariant embedding baseline, even when the latter is trained with additional landmark masking augmentation.

Compared with the baselines, MPE showed a clear and consistent advantage across all masking conditions. V-VIPE was the strongest competing recent single-frame embedding baseline, but it remained consistently below MPE under both full and partial observations. MotionBERT was also weaker, suggesting that although it provides a strong general-purpose pose representation, it is less effective than MPE for fine-grained pose retrieval under the current single-frame retrieval protocol. Pr-VIPE and AE remained relatively competitive in several masked settings—for example, Pr-VIPE reached 0.744 in *Precision@1* under right arm+leg masking, and AE reached 0.744 under left-arm masking—but neither method matched MPE in any metric or masking condition. VAE showed the weakest overall performance, indicating limited robustness for retrieval under both full and partial observations.

Another notable pattern is that masking multiple body parts generally made retrieval more difficult than masking a single limb, as expected, because a larger portion of the pose configuration was removed. Nevertheless, the degradation of MPE was smaller than that of the competing methods. This robustness is particularly important in real-world scenarios, where self-occlusion, truncation, or imperfect pose estimation frequently leads to incomplete landmark observations.

Figure 5 presents representative retrieval examples on Yoga-82. Compared with the baselines, the proposed method retrieves poses that are not only globally similar in silhouette but also more consistent in detailed limb arrangement and overall pose semantics. These qualitative observations are consistent with the quantitative improvements shown in Table 2.

5.3.1 Failure analysis

Although MPE achieved the best overall retrieval performance in Table 2, the remaining errors reveal a clear and interpretable failure mode rather than random mismatches. In particular, retrieval becomes less reliable when severe self-occlusion, compact body configurations, and viewpoint ambiguity occur simultaneously. This observation is also consistent with the controlled correlation analysis in Fig. 4, where the correlation of all methods, including MPE, decreases as viewpoint rotation and landmark masking increase.

Figures 6 and 7 show two representative failure cases. In Fig. 6, corresponding to *Child Pose (Balasana)*, MPE often retrieves poses that share a similar globally folded or forward-bending 2D silhouette with the query, but belong to different semantic categories. In Fig. 7, corresponding to *Reclining Hand-to-Big-Toe Pose (Supta Padangusthasana)*, the retrieved samples frequently preserve a similar leg-extension pattern while confusing the overall body orientation, especially the distinction between supine and prone configurations. These examples indicate that MPE captures coarse structural similarity reliably, but may fail when fine-grained semantic cues are suppressed by projection ambiguity.

These qualitative examples are also consistent with the quantitative trend in Table 2. Retrieval remains relatively stable when only limited pose information is missing, but degrades more noticeably as more discriminative body cues are removed, because the remaining visible landmarks become less sufficient to disambiguate semantically similar poses.

Importantly, these failure cases suggest that the main limitation of MPE is not a failure to encode pose structure itself, but rather the difficulty of disambiguating poses that become similar after 3D-to-2D projection and partial landmark loss. This limitation is particularly pronounced for compact or folded poses, where multiple joints overlap spatially and depth-dependent differences are largely collapsed in the 2D representation. As a result, the model may retrieve poses with a similar overall skeleton layout while missing subtle but semantically critical differences in body orientation or limb arrangement.

Overall, the failure analysis supports the main quantitative findings from both the correlation analysis and the retrieval experiments. MPE remains substantially more robust than the baselines, but its errors concentrate in cases where viewpoint ambiguity and self-occlusion jointly reduce the discriminability of fine-grained pose semantics. These observations point to promising future directions, such as incorporating confidence-aware landmark weighting, explicit orientation-sensitive supervision, or



Fig. 5 Examples of pose retrieval results on the Yoga-82 dataset

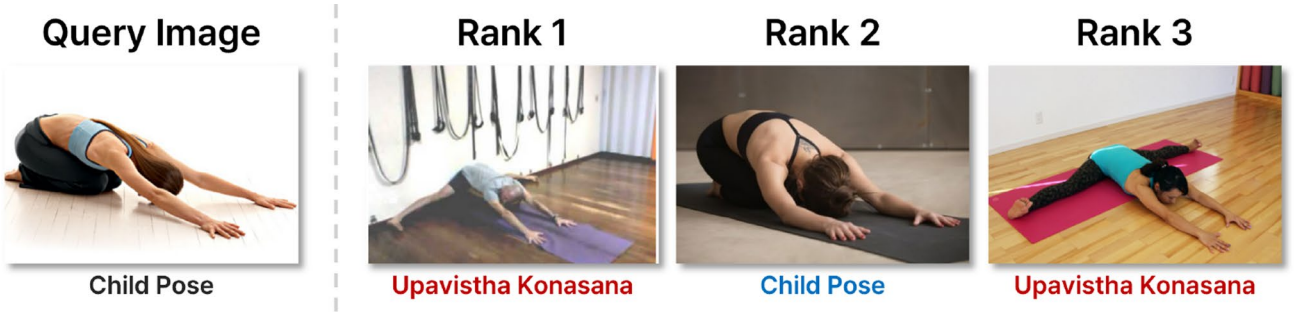


Fig. 6 Failure case on *Child Pose (Balasana)*. Correct matches are highlighted in blue and incorrect matches in red. MPE preserves the coarse folded-pose structure but may confuse categories when compact body configurations and self-occlusion obscure fine-grained cues

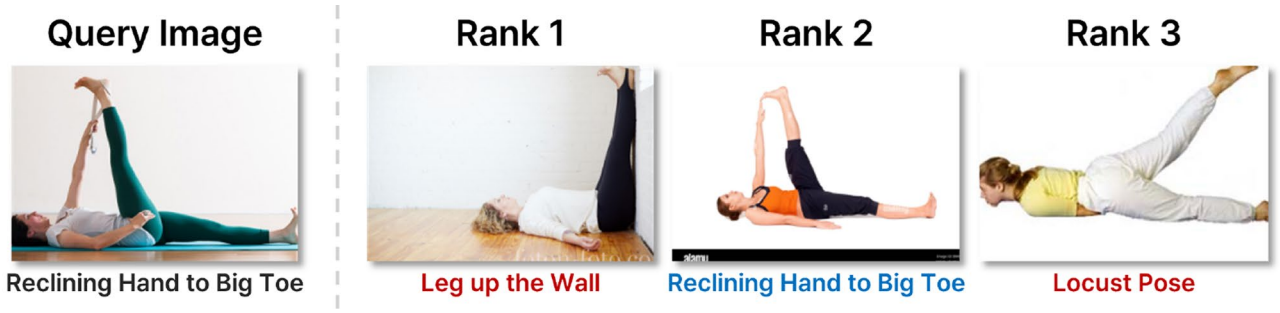


Fig. 7 Failure case on *Reclining Hand-to-Big-Toe Pose (Supta Padangusthasana)*. Correct matches are highlighted in blue and incorrect matches in red. MPE often preserves leg-extension patterns but may confuse global body orientation under 2D projection ambiguity

weak depth-aware constraints to better separate poses with similar projected 2D structures but different underlying 3D semantics.

6 Conclusion

In this paper, we proposed Masked Pose Embedding (MPE), a transformer-based autoencoder for learning view- and occlusion-robust pose embeddings from sparse 2D landmarks. By combining view-angle canonicalization with landmark-level masking and masked-only supervision, MPE preserves 3D-aware pose semantics without relying on contrastive pair mining or variational regularization. This masked-reconstruction formulation provides a training-stable alternative for robust 2D pose embedding.

We evaluated MPE from two complementary perspectives. In the embedding-space analysis, MPE achieved the strongest correlation with ground-truth 3D pose distances among the compared methods under both controlled perturbations on SARA and natural pose distributions on Human3.6M. These results indicate that the learned representation preserves 3D pose semantics more faithfully than the compared baselines, including the V-VIPE baseline. In the downstream pose retrieval task on Yoga-82, MPE also achieved the strongest retrieval performance among the compared methods under both full and partial landmark observations, demonstrating robustness to viewpoint variation and partial occlusion in real image-based scenarios.

An important practical advantage of MPE is its parameter efficiency. Despite using only 1.98M parameters, MPE outperformed the compared recent single-frame V-VIPE baseline and the substantially larger temporal representation baseline MotionBERT (42.47M parameters) in our evaluation settings. This result suggests that robust and semantically meaningful 2D pose embeddings can be learned without necessarily relying on large-scale motion representation models.

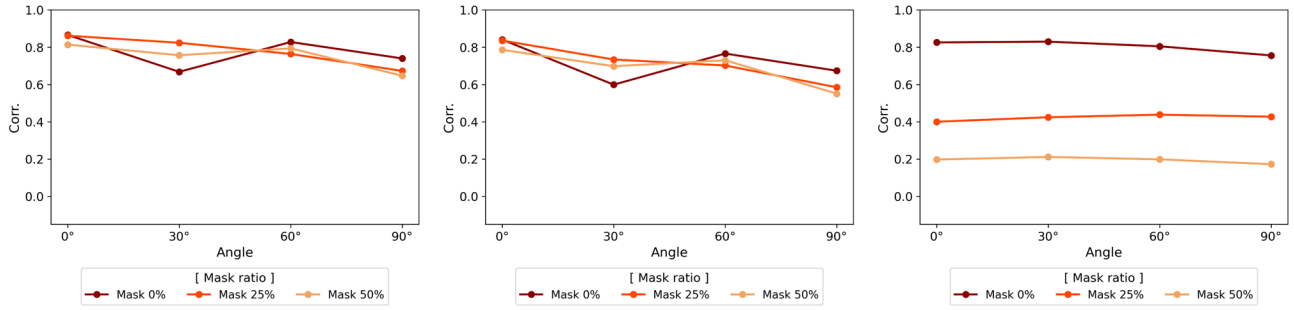
At the same time, our qualitative failure analysis showed that MPE remains challenged in cases where severe self-occlusion and large viewpoint changes occur simultaneously. In particular, compact or folded poses can become difficult to distinguish when depth information is lost through 3D-to-2D projection, leading to confusion between poses with similar projected structures. These observations are consistent with the degradation patterns observed in the quantitative analysis and highlight an important direction for future work.

Another limitation is that MPE uses 3D pose data during training to construct rotated 2D inputs and canonical reconstruction targets, although only 2D landmarks are required at inference time. In addition, the Yoga-82 retrieval evaluation depends on landmarks extracted by YOLOv8x-pose, so errors from the upstream pose detector may affect the final retrieval performance. Finally, the synthetic masking patterns used in our evaluation provide controlled tests of partial visibility, but they may not fully capture the diversity of real-world occlusion, truncation, and detector failure patterns.

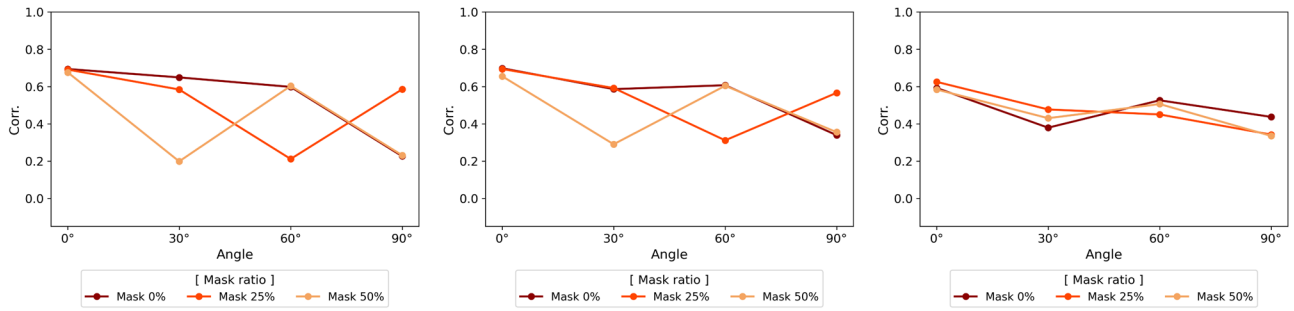
Overall, the proposed MPE offers an effective and compact solution for pose similarity analysis and retrieval from 2D landmarks within the evaluated settings. We believe that this framework can serve as a practical component for downstream applications such as rehabilitation guidance, sports analysis, and pose-based retrieval systems, and can be further extended to handle more extreme viewpoint ambiguity, stronger self-occlusion, and broader in-the-wild pose distributions.

Appendix 1: Alternative visualization of correlation results

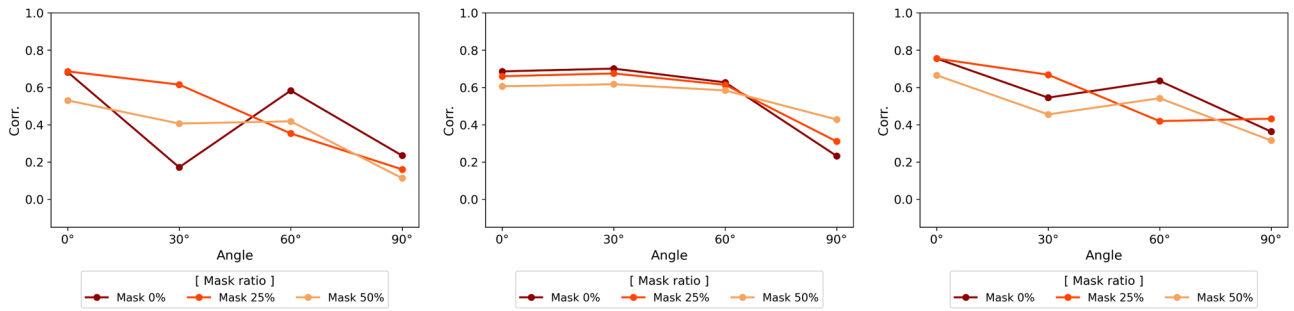
See Fig. 8.



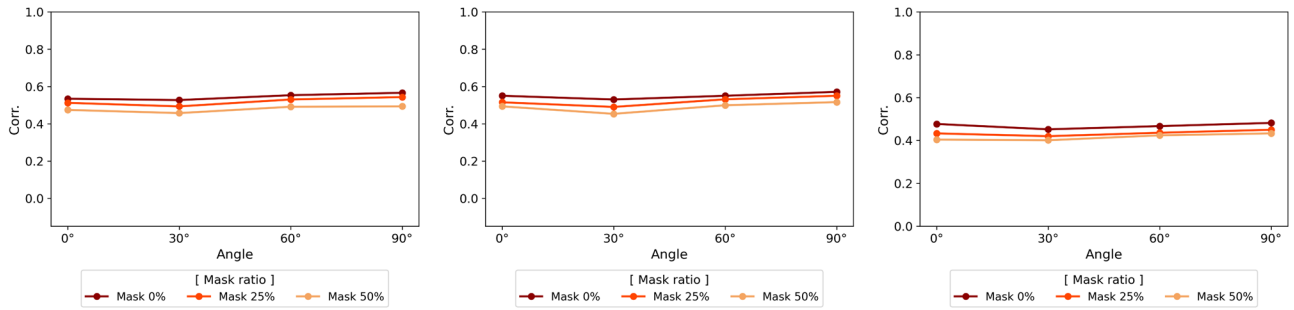
(a) MPE (50% masking) (b) MPE (75% masking) (c) MotionBERT (Original)



(d) Pr-VIPE (Original) (e) Pr-VIPE (Best) (f) VAE (Original)



(g) VAE (Best) (h) AE (Original) (i) AE (Best)



(j) V-VIPE (25% masking) (k) V-VIPE (50% masking) (l) V-VIPE (75% masking)

Fig. 8 Alternative line-plot visualization of Fig. 4. The plots show Pearson correlation across yaw rotations and masking ratios, highlighting degradation trends under increasing viewpoint change and occlusion

Appendix 2: MPE ablation

Appendix 2.1: Training objective ablation

Table 3 Training objective ablation for MPE with a fixed 50% masking ratio. Pearson correlation is reported for masked-only supervision and all-joint supervision under controlled yaw rotation and occlusion conditions

MPE (50% Mask)	(a) Masked-only	(b) All joints
(1) 0° + 0%	.866	**.867**
(2) 0° + 25%	.862	.862
(3) 0° + 50%	**.814**	.807
(4) 30° + 0%	**.668**	.618
(5) 30° + 25%	**.824**	.815
(6) 30° + 50%	.757	**.773**
(7) 60° + 0%	**.828**	.814
(8) 60° + 25%	.764	**.773**
(9) 60° + 50%	**.794**	.767
(10) 90° + 0%	**.740**	.731
(11) 90° + 25%	**.673**	.611
(12) 90° + 50%	**.647**	.599

Bold values indicate the best-performing configuration for each evaluation condition

Appendix 2.2: Masked landmark augmentation ablation

Table 4 Masking-ratio ablation for MPE under masked-only supervision. Pearson correlation is reported for no masking, 50% masking, and 75% masking across controlled viewpoint and occlusion conditions

	(a) w/o mask	(b) 50% mask	(c) 75% mask
(1) 0° + 0%	.777	**.866**	.841
(2) 0° + 25%	.744	**.862**	.836
(3) 0° + 50%	.683	**.814**	.786
(4) 30° + 0%	.610	**.668**	.599
(5) 30° + 25%	.336	**.824**	.734
(6) 30° + 50%	.221	**.757**	.698
(7) 60° + 0%	.331	**.828**	.766
(8) 60° + 25%	.223	**.764**	.702
(9) 60° + 50%	.334	**.794**	.730
(10) 90° + 0%	.226	**.740**	.674
(11) 90° + 25%	.336	**.673**	.585
(12) 90° + 50%	.224	**.647**	.551

Bold values indicate the best-performing configuration for each evaluation condition

Appendix 2.3: View-angle-based landmark augmentation ablation

Table 5 View-angle augmentation ablation for MPE. Pearson correlation is reported with and without rotation augmentation under controlled yaw rotation and occlusion conditions

	(a) w/o rotation aug.	(b) w/ rotation aug.
(1) 0° + 0%	.856	**.866**
(2) 0° + 25%	.838	**.862**
(3) 0° + 50%	.791	**.814**
(4) 30° + 0%	**.813**	.668
(5) 30° + 25%	.797	**.824**
(6) 30° + 50%	.755	**.757**
(7) 60° + 0%	.726	**.828**
(8) 60° + 25%	.718	**.764**
(9) 60° + 50%	.689	**.794**
(10) 90° + 0%	.602	**.740**
(11) 90° + 25%	.609	**.673**
(12) 90° + 50%	.600	**.647**

Bold values indicate the best-performing configuration for each evaluation condition

Appendix 2.4: Architecture ablation

For architecture selection, we used the mean Pearson correlation across the 12 controlled SARA evaluation conditions as the aggregate criterion. The selected configuration achieved the highest aggregate mean among the tested variants, although some alternatives, particularly dim=64, were competitive and outperformed it under specific individual conditions. We therefore report the full condition-wise results below to show both the aggregate trend and the condition-specific trade-offs.

Table 6 Effect of encoder depth

	(a) Enc=3	(b) Enc=6	(c) Enc=9
(1) 0° + 0%	.778	**.866**	.815
(2) 0° + 25%	.757	**.862**	.803
(3) 0° + 50%	.693	**.814**	.769
(4) 30° + 0%	.767	.668	**.794**
(5) 30° + 25%	.747	**.824**	.786
(6) 30° + 50%	.685	**.757**	.752
(7) 60° + 0%	.721	**.828**	.728
(8) 60° + 25%	.701	**.764**	.731
(9) 60° + 50%	.645	**.794**	.705
(10) 90° + 0%	.585	**.740**	.567
(11) 90° + 25%	.579	**.673**	.599
(12) 90° + 50%	.547	**.647**	.602

Bold values indicate the best-performing configuration for each evaluation condition

Table 7 Effect of embedding dimension

	(a) dim=64	(b) dim=128	(c) dim=256
(1) 0° + 0%	.835	**.866**	.730
(2) 0° + 25%	.823	**.862**	.705
(3) 0° + 50%	.796	**.814**	.629
(4) 30° + 0%	**.818**	.668	.714
(5) 30° + 25%	.809	**.824**	.692
(6) 30° + 50%	**.784**	.757	.618
(7) 60° + 0%	.774	**.828**	.646
(8) 60° + 25%	**.771**	.764	.633
(9) 60° + 50%	.753	**.794**	.566
(10) 90° + 0%	.670	**.740**	.465
(11) 90° + 25%	**.685**	.673	.467
(12) 90° + 50%	**.685**	.647	.434

Bold values indicate the best-performing configuration for each evaluation condition

Table 8 Effect of the number of attention heads

	(a) head=2	(b) head=4	(c) head=8
(1) 0° + 0%	.761	**.866**	.727
(2) 0° + 25%	.735	**.862**	.730
(3) 0° + 50%	.686	**.814**	.738
(4) 30° + 0%	**.745**	.668	.705
(5) 30° + 25%	.720	**.824**	.707
(6) 30° + 50%	.673	**.757**	.716
(7) 60° + 0%	.677	**.828**	.631
(8) 60° + 25%	.660	**.764**	.639
(9) 60° + 50%	.626	**.794**	.656
(10) 90° + 0%	.516	**.740**	.460
(11) 90° + 25%	.525	**.673**	.478
(12) 90° + 50%	.523	**.647**	.517

Bold values indicate the best-performing configuration for each evaluation condition

Table 9 Effect of decoder depth

	(a) dec=1	(b) dec=2	(c) dec=4
(1) 0° + 0%	.819	**.866**	.823
(2) 0° + 25%	.801	**.862**	.811
(3) 0° + 50%	.740	**.814**	.768
(4) 30° + 0%	.804	.668	**.805**
(5) 30° + 25%	.788	**.824**	.793
(6) 30° + 50%	.726	**.757**	.751
(7) 60° + 0%	.742	**.828**	.746
(8) 60° + 25%	.735	**.764**	.741
(9) 60° + 50%	.677	**.794**	.710
(10) 90° + 0%	.575	**.740**	.601
(11) 90° + 25%	.589	**.673**	.629
(12) 90° + 50%	.551	**.647**	.632

Bold values indicate the best-performing configuration for each evaluation condition

Appendix 3: Baseline configuration ablation

Tables 10, 11, 12, and 13 report additional baseline configuration ablations under controlled viewpoint and occlusion conditions.

Table 10 Baseline configuration ablation for VAE. Pearson correlation is reported for masking with rotation augmentation, masking only, and rotation augmentation only

	(a) Mask+rot.	(b) Only mask	(c) Only rot.
(1) 0 + 0%	.592	.682	.625
(2) 0 + 25%	.625	.686	.625
(3) 0 + 50%	.584	.530	.606
(4) 30 + 0%	.379	.171	.422
(5) 30 + 25%	.477	.615	.477
(6) 30 + 50%	.430	.406	.277
(7) 60 + 0%	.526	.583	.487
(8) 60 + 25%	.450	.353	.291
(9) 60 + 50%	.506	.418	.466
(10) 90 + 0%	.437	.234	.274
(11) 90 + 25%	.342	.160	.361
(12) 90 + 50%	.335	.113	.199

Table 11 Baseline configuration ablation for AE. Pearson correlation is reported for masking with rotation augmentation, masking only, and rotation augmentation only

	(a) Mask+rot.	(b) Only mask	(c) Only rot.
(1) 0 + 0%	.686	.755	.741
(2) 0 + 25%	.701	.755	.725
(3) 0 + 50%	.627	.665	.634
(4) 30 + 0%	.232	.545	.438
(5) 30 + 25%	.660	.668	.511
(6) 30 + 50%	.606	.455	.321
(7) 60 + 0%	.675	.635	.542
(8) 60 + 25%	.617	.419	.360
(9) 60 + 50%	.614	.542	.518
(10) 90 + 0%	.584	.363	.379
(11) 90 + 25%	.310	.432	.394
(12) 90 + 50%	.428	.315	.325

Table 12 Baseline configuration ablation for Pr-VIPE. Pearson correlation is reported for different combinations of masking ratio and triplet margin

	(a) Mask=20% (margin=2)	(b) Mask=20% (margin=3)	(c) Mask=50% (margin=2)	(d) Mask=50% (margin=3)
(1) 0 + 0%	.694	.698	.634	.692
(2) 0 + 25%	.691	.694	.627	.685
(3) 0 + 50%	.676	.655	.606	.644
(4) 30 + 0%	.649	.586	.576	.565
(5) 30 + 25%	.584	.592	.300	.488
(6) 30 + 50%	.198	.290	.190	.284
(7) 60 + 0%	.598	.607	.296	.481
(8) 60 + 25%	.211	.311	.194	.278
(9) 60 + 50%	.603	.604	.283	.449
(10) 90 + 0%	.226	.339	.195	.251
(11) 90 + 25%	.585	.566	.263	.390
(12) 90 + 50%	.231	.355	.189	.202

Table 13 V-VIPE masking-ratio ablation. Pearson correlation is reported for 25%, 50%, and 75% training masking ratios under controlled viewpoint and occlusion conditions

	Mask=25%	Mask=50%	Mask=75%
(1) 0 + 0%	.534	.550	.477
(2) 0 + 25%	.512	.515	.433
(3) 0 + 50%	.474	.493	.404
(4) 30 + 0%	.527	.530	.452
(5) 30 + 25%	.493	.490	.420
(6) 30 + 50%	.457	.453	.401
(7) 60 + 0%	.553	.550	.467
(8) 60 + 25%	.530	.531	.436
(9) 60 + 50%	.491	.499	.424
(10) 90 + 0%	.566	.571	.482
(11) 90 + 25%	.543	.550	.450
(12) 90 + 50%	.493	.516	.433

Author Contributions J.K. conceived the main idea, designed and conducted the experiments, analyzed the results, and wrote the manuscript. J.S. trained and evaluated the models, analyzed the results, and contributed to revise the manuscript. M.K. supervised the project, provided guidance on methodology and analysis, and reviewed the manuscript. All authors reviewed the manuscript.

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Data Availability The SARA dataset is publicly available at https://github.com/chico2121/SARA_Dataset. The Human3.6M dataset is available from the official project website at https://vision.imar.ro/human3.6m/, subject to its data usage and citation requirements. The Yoga-82 dataset is publicly available at https://sites.google.com/view/yoga-82/home#h.whml2duhj9wa.

Declarations

Conflict of interest The authors declare no conflict of interest.

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